Antimicrobial Resistance in Bulgaria: A Policy Feedback Analysis of the 2023 eRx Mandate

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Contents

Abstract

Antimicrobial resistance (AMR) is a global health crisis that undermines modern medicine, causes hundreds of thousands of deaths annually, and threatens economic stability and food security worldwide. Bulgaria exhibits among the highest bacterial resistance rates in the EU, particularly for Escherichia coli resistance to third-generation cephalosporins, driven by a decade of rising antibiotic consumption that defies the broader European trend. This study applies policy feedback theory and historical institutionalism to identify which systemic determinants – institutional capacity, socioeconomic barriers, or behavioral norms – most strongly explain Bulgaria's persistent crisis. Using domain-specific OLS time-series regressions (2006-2023), EARS-Net resistance data is analyzed alongside indicators spanning public health expenditure, healthcare personnel density, and government effectiveness; Gini coefficient and out-of-pocket payments; and corruption perceptions and rule of law. Governance quality emerges as the strongest predictor, with the rule of law index showing the largest negative association with resistance. The paper then applies Ostrom's rules-in-form versus rules-in-use distinction and the ADICO institutional grammar to diagnose where the 2023 mandatory electronic prescribing mandate is structurally vulnerable, finding that ambiguous penalty provisions and dispersed enforcement authority risk reproducing the informal bargaining dynamics the reform is intended to displace. By grounding the analysis in historically constructed institutional pathways rather than consumption volume alone, the paper offers targeted governance recommendations applicable to other high-burden contexts facing similar structural constraints.

Keywords: antimicrobial resistance, public health governance, policy feedback, institutional capacity, socioeconomic determinants, Bulgaria

Abbreviations alphabetized

AMR	Antimicrobial resistance
BDA	Bulgarian Drug Agency
EARS-Net	European Antimicrobial Resistance Surveillance Network
EC	European Commission
ECDC	European Centre for Disease Prevention and Control
EEA	European Economic Area
eRx	Bulgaria’s mandatory electronic prescription system
ESAC-Net	European Surveillance of Antimicrobial Consumption Network
EPHA	European Public Health Alliance
EU	European Union
GLASS	WHO Global Antimicrobial Resistance and Use Surveillance System
MoH	Ministry of Health
MPHMA	Medicinal Products in Human Medicine Act
NHIS	National Health Information System
NSI	National Statistical Institute
OECD	Organisation for Economic Co-operation and Development
RHI	Regional Health Inspectorates
WHO	World Health Ornagization
WOAH	World Organisation for Animal Health
3GC	Third-generation cephalosporins (class of broad-spectrum antibiotics including ceftriaxone, cefotaxime, ceftazidime)
CPI	Corruption Perceptions Index

I. Introduction

The discovery and development of antimicrobial drugs in the 20th century allowed for effective treatment of a myriad of common widespread infections, saved millions of lives, increased life expectancy, and became a cornerstone of modern medicine. Since then, antimicrobial resistance (AMR) has become one of the most pressing health concerns globally. Excessive or inappropriate use of such treatments has become a common trend in recent years, posing unnecessary selective pressure and promoting adaptation or mutation, which can produce strains resistant to existing drugs (Habboush & Guzman, 2023). The prevalence of the issue has been recognized around the world, and there have been significant efforts to combat the emergence of increasing rates of resistant microbial strains. In 2015, the World Health Organization (WHO) established the Global Antimicrobial Resistance and Use Surveillance System (GLASS), tasked with collecting data and tracking the progression of AMR in 93 infection-pathogen-drug common combinations as an instrument to guide public health policy. Their latest report utilizing data from 2023 indicates that, while participation has increased by 110 countries since its launch, the global trends seem grim, with roughly one in six bacterial infections worldwide showing resistance (WHO, 2025a). Specifically, the most alarming findings indicate that Approximately 40% of *E. coli* and 55% of *K. pneumoniae* bloodstream infections, two of the most common causes of sepsis worldwide, are now resistant to third-generation cephalosporins (3GC) like ceftriaxone, their first-choice treatment (WHO, 2025b).

The immediate health-related effects are that hospital stays are becoming longer and death rates increase. In fact, a report published in 2020 estimates that more than 35 000 deaths are directly caused by AMR in the European Economic area (EEA) according to the European

Centre for Disease Prevention and Control (ECDC) data in 2024. The emergence of resistant microbes will eventually undermine the safety of critical medical interventions, including surgery, chemotherapy, organ transplantation, and intensive care, all of which depend on reliable infection prevention. Such procedures will simply become too dangerous to undertake. Antimicrobials are also heavily used in agriculture and livestock, driving resistance that threatens both human health and food security. By 2050, AMR could cut global GDP by up to 3.8% annually. Low- and middle-income populations bear the heaviest burden due to poor healthcare access, higher infection rates, and counterfeit drugs, deepening inequality (World Bank Group, 2024). The resulting economic disruption is likely to deepen inequality, placing the heaviest burden on populations with the least financial resilience.

In response to growing evidence of AMR's impact on health systems and economies, the EU and its agencies have placed resistance control at the forefront of their public health agenda, as indicated by the European Commission (Directorate-General for Health and Food Safety, 2023). The One Health Framework, endorsed by the WHO, the World Organisation for Animal Health (WOAH), and the Food and Agriculture Organization (FAO), serves as a directive that recognizes that the health of humans, animals, and ecosystems are interdependent, and provides data-backed best-practices. It emphasizes the need for coordinated action across domains and across EU Member States, which are expected to align their national AMR strategies with One Health principles (OECD Health Policy Studies, 2023). Furthermore, Directive 2001/83/EC of the European Parliament clearly necessitates that, in the EU, all antibiotics for human use must be sold only upon presenting a valid prescription (European Commission, 2017).

While AMR arises from diverse biological, behavioral, and environmental drivers, ranging from bacterial mutations and agricultural overuse to self-medication and poor infection control, this study focuses specifically on human antibiotic use patterns in Bulgaria. Clinicians' reliance on experience and pharmaceutical influence, as noted above, exemplifies governance and behavioral gaps that sustain resistance beyond mere consumption volume. Critically, it evaluates whether the 2023 eRx mandate, which is intended to curb such misuse, will prove effective given these systemic constraints. Drawing on Al-Khalaifah et al. (2025) and Irfan et al. (2022), Table 1 presents a comprehensive categorization of the broad landscape of AMR drivers across anthropogenic transmission pathways.

Table 1. Key AMR Drivers and Study Scope

Driver category	**Examples**	**Mechanism**
Human (primary focus)	Improper prescribing; self-medication in human healthcare; improper disposal of hospital wastewater	Clinical overuse of broad-spectrum antibiotics generating selective pressure favoring resistant strains (Irfan et al., 2022) that spread nosocomially and via hospital wastewater (Al-Khalaifah et al., 2025)
Veterinary and livestock	Improper use and overuse similar as in human medicine	Routine growth promotion/prophylaxis in animal farming; Zoonotic transfer via meat/food chain (Al-Khalaifah et al., 2025)
Agricultural	Manure runoff, contaminated irrigation water	Crop exposure to resistant bacteria and spread via farm waste (Al-Khalaifah et al., 2025)
Environmental	WWTP effluents, soil/water reservoirs containing resistant strains	Resistant bacteria contaminate wastewater, rivers, and soil, forming persistent environmental reservoirs that continuously re-infect humans and animals (Irfan et al., 2022)

While biological mechanisms govern resistance emergence, the distinction between intrinsic microbial processes and anthropogenic drivers proves analytically crucial. Resistance is inherently a biological phenomenon, arising from evolutionary interactions between microorganisms and the microbial drugs targeting them. Human, veterinary, agricultural, and environmental drivers determine its transmission and prevalence across populations, and targeted interventions can meaningfully slow this spread: curbing human misuse reduces selective pressure, while regulating veterinary/agricultural inputs limits spillover reservoirs. This study examines inappropriate human antibiotic use patterns in Bulgaria, specifically evaluating the extent to which governance interventions influence national resistance rates.

Over the past decade, Bulgaria has been one of the few EU countries showing a statistically significant increasing trend in total antibacterial consumption in both the community and hospital sectors, in contrast to the generally decreasing EEA average. European Surveillance of Antimicrobial Consumption Network (ESAC-Net) metrics show standardized outpatient antibiotic prescriptions (ATC J01) as daily doses per 1,000 population, revealing Bulgaria's uniquely rising trend. At 26.7 DDD/1,000 inh/day (2023), Bulgaria tripled the lowest performers like the Netherlands (8.8) and exceeded the EEA mean (~20.0) by 33%. Notably, broad-spectrum agents dominate >60% of use and continue rising, accelerating resistance pressure (ECDC, 2024). The same report published the previous year flags Bulgaria as the one country with a constant increase in consumption levels observed for both community/outpatient and inpatient sectors for over a decade of data collection (ECDC, 2023). The combination of Bulgaria's uniquely persistent upward trend in consumption, levels significantly exceeding both the EEA mean and responsible-prescribing benchmarks such as the Netherlands, and the dominance of broad-spectrum agents beyond clinically justified thresholds collectively indicates that antibiotic use in Bulgaria extends well beyond genuine epidemiological need, meeting the established definition of overuse and inappropriate prescribing (WHO, 2016).

In fact, some of the main reasons for the high consumption levels were revealed nearly a decade ago during a One Health Country Visit, aimed at assisting the development of country-specific strategies to reach EU goals, where ECDC and and the European Commission's Directorate-General for Health and Food Safety jointly outline key areas in need of improvement. The recommendations include limiting blind prescribing of broad-spectrum antibiotics without microbiological testing, absence of national treatment guidelines, and reliance on personal judgment, colleague advice, or pharmaceutical company information (ECDC, 2016).

Despite sustained empirical evidence positioning Bulgaria among high-consumption outliers, the corresponding legislative response has been slow to materialise. Mandatory electronic prescribing (eRx) for antibiotics was only introduced within Bulgaria's existing legal framework in 2023 by the Ministry of Health, specifically in Ordinance No. 4 under the Medicinal Products in Human Medicine Act (MPHMA), and fully enforced from April 2024, under the authority of the Minister of Health and the Council of Ministers (Министерство на здравеопазването, 2024). Among the advantages of this first meaningful step toward tightening regulations on antibiotic use are transparency and reliable data collection as each purchase requires a valid diagnosis with traceable information about the prescriber, dispenser, and patient. Compliance monitoring is carried out jointly by the Regional Health Inspectorates (RHI) and the Bulgarian Drug Agency (BDA), the latter holding authority to impose fines, sanctions, and permit cancellations in cases of infringement of the MPHMA (BDA, 2021), with specific infringing provisions outlined in Appendix A. Authorities have credited the transition with substantial reductions in uncontrolled antibiotic use (Information Services, 2025). However, emerging evidence suggests that the intervention has so far fallen short of producing statistically demonstrable reductions in overall consumption (Vankova et al., 2025). This may be partly attributable to the recency of full enforcement, as the policy has been operational for too limited a period to generate the longitudinal data necessary for robust statistical inference, and behavioural change at the prescriber and patient level can reasonably be expected to lag behind regulatory implementation. It is within this context that the present study is situated, with the primary objective of evaluating the potential of mandatory eRx to meaningfully influence antibiotic consumption in Bulgaria. To that end, the study will examine both the regulatory conditions that necessitated the adoption of this policy, and the foreseeable implications of its continued implementation for antimicrobial stewardship and public health outcomes.

II. Problem Definition

Antibiotic resistance is not a distant or theoretical threat, but a measurable, accelerating crisis with direct consequences for the clinical and economic sustainability of health systems. To understand why it is so difficult to contain, it is necessary to first situate it within the broader landscape of antimicrobial medicine. Antimicrobials are a broad class of therapeutic agents designed to inhibit or eliminate microorganisms, and they are typically grouped by the type of organism they target: antibiotics act against bacteria, antifungals against fungi, and antivirals against viruses (Microbiology Society, 2020). Resistance, or the ability of a microorganism to survive exposure to a drug that would ordinarily kill it, can develop across all three classes, and each presents its own public health challenges. However, it is antibiotic resistance that constitutes the most immediate and well-documented threat to human health globally, and it is the primary driver of concern in the Bulgarian context, where patterns of human antibiotic use in both community and hospital settings have been flagged as outliers within the EU for over a decade (ECDC, 2024).

Bacteria are among the fastest-reproducing organisms on earth, with large populations and rapid replication cycles that generate frequent spontaneous random mutations. A small fraction of these confer survival advantages under drug exposure – for example, by altering the molecular target an antibiotic binds to, by producing enzymes that break the drug down, by reducing membrane permeability to limit drug entry, or by activating efflux pumps that actively expel the drug before it can act (Belay et al., 2024). The odds of any single bacterium carrying a resistance-conferring mutation are extraordinarily low – between one in a million and one in ten billion per replication cycle – yet given that a typical infection involves populations of billions of cells dividing rapidly, the statistical near-certainty is that resistant variants already exist within the population before a patient ever takes their first dose (Jacoby & Low, 2010). When a population of bacteria is exposed to an antibiotic, susceptible strains are eliminated while those carrying resistance traits survive and multiply, a process of evolutionary selection that is not incidental but inevitable under sufficient drug pressure (Naghavi et al., 2024). Critically, resistance genes can be transferred not only vertically from parent to offspring, but horizontally between entirely different bacterial species via mobile genetic elements such as plasmids and transposons, meaning that resistance acquired in one organism can rapidly disseminate across a broader microbial ecosystem (Salam et al., 2023). Inadequate dosing and incomplete treatment courses are particularly dangerous in this regard, as they expose bacteria to sublethal drug concentrations: enough to exert selective pressure, but insufficient to eliminate the population, effectively accelerating adaptation rather than preventing it (Cox & Wright, 2013).

This biological logic carries a direct policy implication. Every unnecessary prescription, every incomplete treatment course, and every instance of dispensing without clinical indication contributes to the selective pressure under which resistant strains proliferate. Crucially, the consequences do not remain local: resistant strains spread through healthcare networks, communities, and across borders, meaning that even responsible prescribers bear exposure to resistance generated elsewhere (Belay et al., 2024). This is what ultimately makes AMR a collective action problem: the costs of overuse are distributed across entire populations and health systems, while the immediate incentive structures faced by individual prescribers and patients frequently point in the opposite direction.

WHO's latest pipeline review identified only 27 antibiotics in clinical development targeting the most critical drug-resistant priority pathogens, and of those, only six were classified as innovative — meaning they use a genuinely new mechanism of action or target site that resistant bacteria have not yet encountered, and are therefore unlikely to be immediately rendered ineffective by existing resistance (WHO, 2024). The remaining 21 are variations on existing drug classes, meaning resistance to them may already be present or rapidly emerging in bacterial populations. The high cost of antibiotic development, combined with the relatively short period of clinical efficacy before resistance emerges, has led to a sustained decline in pharmaceutical investment that further compounds the crisis (Salam et al., 2023). Beyond conventional drug development, emerging research is exploring alternative strategies including interference with bacterial communication systems (quorum sensing) to disrupt biofilm formation and toxin production (O'Loughlin et al., 2013), and AI-assisted bacteriophage design, where computational models have been used to generate candidate viruses capable of targeting antibiotic-resistant bacteria (Kavanagh, 2025). While such approaches are scientifically promising, they remain largely in experimental stages and are not yet available as clinical tools at scale. A durable solution will ultimately require a breakthrough from the laboratory, whether through new drug classes, novel mechanisms of action, or these alternative therapeutic strategies, but given the depth of the current innovation gap, no such solution is imminent. In the interim, the most effective tool available to health systems is governance: reducing unnecessary consumption, tightening prescribing practices, and slowing the pace at which resistance accumulates.

So why is it that some EU countries make greater progress in controlling antimicrobial resistance than others, given the clear directives issued to all member states? Pierre et al. (2023) demonstrate that executive capacity, encompassing policy execution, coordination, and enforcement, is a stronger predictor of resistance prevalence across EU states than consumption volume alone, with Eastern European countries' common capacity deficits explaining a substantial share of their elevated resistance burdens. AMR is therefore not solely a biomedical issue but a governance challenge: because resistance emerges from countless individual prescribing and dispensing decisions that accumulate into harmful population-level outcomes, well-designed and consistently enforced public policies can meaningfully slow its spread in ways that medical knowledge alone cannot. Where institutional capacity is limited, technical measures such as prescription-only requirements or surveillance systems are difficult to implement consistently, and as Collignon et al. (2015) show, governance quality, particularly corruption control, explains as much variation in resistance rates across European countries as antibiotic consumption itself. This work aims to identify the specific governance weaknesses that sustain inappropriate antibiotic use in Bulgaria and evaluate whether the 2023 eRx mandate is designed to address them.

III. Literature Review

This literature review adopts Policy Feedback Theory (PFT) as an organizing framework to examine how past policy decisions in Bulgaria's healthcare governance have shaped the structural conditions driving high antibiotic consumption and, ultimately, necessitated the introduction of mandatory electronic prescribing. PFT holds that public policies, once enacted, do not merely respond to political conditions but actively reshape them, becoming causes of subsequent political and social outcomes in their own right (Pierson, 1993). Pierson's foundational argument is that policies generate resources and incentives for political actors, and provide those actors with information and cues that encourage particular interpretations of the political world, mechanisms that operate on government elites, interest groups, and mass publics alike. Semantic Scholar Critically, these feedback effects can generate path-dependent dynamics: self-reinforcing feedbacks create benefits, increase the cost of policy change, and produce path dependence, while self-undermining feedbacks erode policy legitimacy and create pressure for reform.

Mettler and Soss (2004) extend this framework to show how policy designs define beneficiaries, structure authority relations, and shape citizens' orientations toward government — with consequences that extend well beyond the immediate policy goals. Crucially, PFT is primarily a theory of political and policy dynamics: it explains how past policy choices create constituencies, institutional routines, and normative expectations that make future reform harder or easier, not merely how any social phenomenon produces societal effects. Applied to the Bulgarian AMR context, this framework enables identification of the specific policy-feedback pathways – institutional, socioeconomic, and normative – through which earlier governance decisions have entrenched patterns of inappropriate antibiotic use, created the conditions necessitating the eRx mandate, and will shape its prospects for success. Drawing on the resource/interpretive distinction developed by Pierson (1993) and elaborated by Mettler and Soss (2004), the analysis organizes these pathways into three domains: institutional capacity, socioeconomic barriers, and behavioral norms. Together, these operationalize the feedback mechanisms through which past policy legacies constrain current reform efforts, and through which the eRx intervention may, or may not, generate new self-reinforcing dynamics.

Institutional Capacity

A substantial body of research identifies governance failures as the primary driver of antimicrobial resistance, demonstrating how past policy choices create persistent weaknesses in regulatory capacity, enforcement, and surveillance through resource effect feedbacks. Factors such as organizational culture, workforce management, and interorganizational collaboration significantly shape healthcare outcomes (Bhat et al., 2022). Integrated healthcare systems particularly depend on effective coordination of these elements to manage complex needs. In Bulgaria, fragmented authority between the Bulgarian Drug Agency (BDA) for national licensing/inspections, Regional Health Inspectorates (RHI) for local enforcement, and Ministry of Health for policy creates coordination failures (ECDC, 2016). Only 30% of hospitals implement antimicrobial stewardship programs, microbiology laboratory capacity serves less than 50% of the population, 45% of antibiotics were sold over-the-counter pre-eRx, and no pharmacy sales monitoring existed (WHO, 2022). Weak interorganizational collaboration obstructs consistent regulation across regions, limiting monitoring and penalty effectiveness.

Pierre & Pierre (2024) demonstrate that executive capacity, encompassing policy execution, coordination, and learning, significantly predicts lower resistance prevalence across EU states, with a 1-unit capacity increase (1-10 scale) linked to 19% lower *E. coli* resistance after controls. Their analysis includes a discussion on how Eastern Europe's common capacity deficits can explain elevated resistance burdens beyond consumption volume alone. Bulgaria consistently ranks lowest on these metrics, correlating with its outlier consumption. WHO Europe (2023) links surveillance underfunding to patient safety failures, noting Bulgaria's gaps enable undetected resistance outbreaks. OECD (2023) highlights Bulgaria's absent One Health governance body, creating human-veterinary silos responsible for 40% resistance spillover. ECDC's 2019 country visit identified Bulgaria's lack of coordinating mechanism and separate sectoral plans as primary institutional barriers. Vankova et al. (2025) document e-Rx enforcement gaps such as workflow delays, absent diagnostic rationale, rigid error handling, outdated NHIS listings, complex searches, system downtimes, and incomplete medical coverage, potentially explaining the policy's limited short-term impact on consumption and revealing systemic inefficiencies.

These institutional deficiencies create self-reinforcing resource effect loops: weak surveillance enables unchecked antibiotic consumption, which drives rising AMR and further erodes already strained capacities. Historical policy choices, namely the 1990s shift to insurance-based financing alongside MPHMA's optional stewardship provisions, entrenched these gaps, leaving execution chronically fragile (World Bank, 2022). The empirical analysis will test whether institutional capacity exhibits negative association with AMR rates, building on Pierre & Pierre (2024). This relationship will be examined through domain-specific fixed-effects regressions on EARS-Net data, comparing standardized coefficients across capacity indicators (health expenditure, personnel density, government effectiveness).

Socioeconomic Barriers

Persistent resource constraints, staffing shortages, and inadequate training in LMICs degrade care quality, often resulting in substandard treatment, medical errors, and poor patient safety outcomes (Kumah et al., 2025). Empirical studies further highlight socioeconomic conditions and healthcare access as strong predictors of population health (Hood et al., 2016). Globalization, mainly through global market integration and neoliberal policies, reshapes employment conditions, income distribution, and access to social safety nets, ultimately exacerbating health inequities within and across countries. Labonté et al. (2011) highlight the growing challenges of effective healthcare governance in an increasingly globalized world—one where health threats such as pandemics and unsafe product flows are intensifying, while trade agreements and financial volatility constrain national policy space. They argue that public health practitioners must adopt transnational strategies, stronger surveillance, and active advocacy to respond to these pressures. Macroeconomic conditions can therefore also be used as determinants of health system performance across populations, with economic growth, fiscal stability, and employment levels influencing the resources governments can allocate to healthcare, the capacity to invest in infrastructure and workforce development, and the resilience of health systems during shocks.

This insight aligns with the Social Determinants of Health (SDOH) perspective, which emphasizes that political, economic, and social structures create the conditions in which people live and access care. Solar & Irwin (2010) present the WHO CSDH conceptual framework distinguishing structural determinants (governance, macroeconomic/social policies, societal values) that generate socioeconomic position (income, education, power) from intermediary determinants (material conditions, behaviors, health system access) driving health inequities. Applying this framework to antimicrobial resistance highlights that patterns of antibiotic use are not simply the result of individual choices or clinical decisions, but emerge from the broader structural context, such as regulatory quality, institutional trust, economic insecurity, and health-system capacity, that shapes how patients seek care and how clinicians prescribe. In other words, understanding AMR in Bulgaria requires examining how these structural and intermediary determinants interact to produce the behaviors and system failures that drive resistance.

In the context of Bulgaria, patients bear a disproportionately large portion of healthcare costs directly, with out-of-pocket payments making up 34% of total spending – more than double the EU average (WHO, 2023) – which further enhances existing inequality in regional access to healthcare facilities and education. While access to healthcare is legally guaranteed in Bulgaria, in practice significant disparities exist, especially for low-income and members of marginalized groups (Caradja & Ivanova, 2015). These socioeconomic barriers, combined with systemic weaknesses in governance and institutional capacity, help explain the patterns of inappropriate antibiotic use and high AMR rates observed in the country, highlighting the need to address both structural and behavioral determinants.

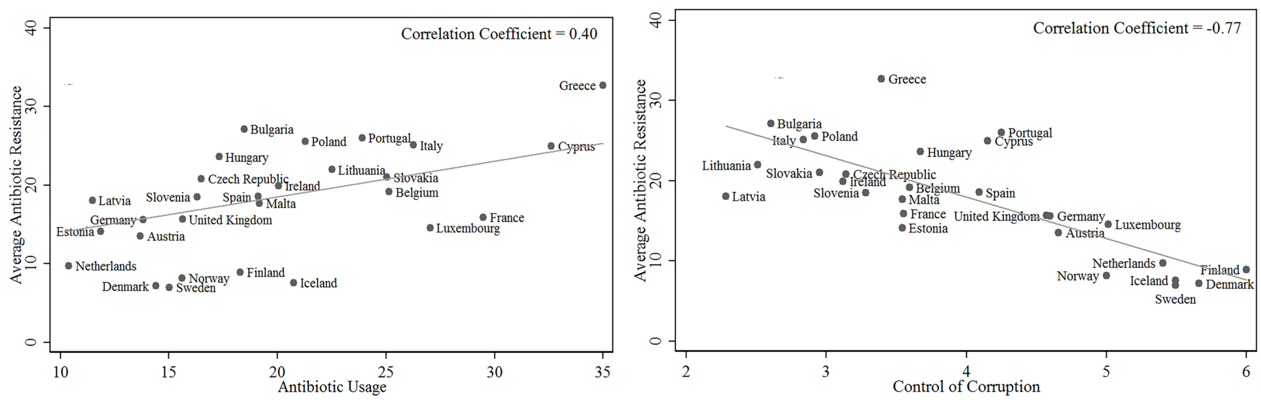
Behavioral Norms

Policy feedback theory identifies interpretive effects as mechanisms through which policies convey meanings, frame problems, and ultimately affect citizens' understandings of government competence and appropriate behaviors. Low trust in health authorities undermines compliance, driving self-directed actions that bypass formal systems. Systematic reviews across chronic diseases show distrust doubles non-adherence rates and serves as a reliable predictor of self-medication across contexts, as patients perceive formal care as unreliable (Bazargan et al., 2021). A cross-national survey (~50,000 respondents spanning 23 countries) found consistent evidence showing that higher trust in government consistently predicted greater engagement in preventive health behaviors (handwashing, avoiding crowds, quarantine) and prosocial actions (donating, volunteering) during COVID-19 (Han et al., 2021). Trust also slowed declines in these behaviors over time, especially in middle-income countries. Key drivers of trust in the study included government organization, perceived fairness, clear messaging, and knowledge, highlighting that transparent, equitable policies can sustain compliance and prosocial behavior.

How trust in healthcare institutions affects vaccinations:

Cultural attitudes in Bulgaria toward antibiotic use, characterized by widespread self-medication and tolerance for obtaining antibiotics without prescription, provide insights into the inefficiencies in eRx policy outcomes. Bulgarians exhibit strong tendencies toward self-use, defined by WHO as individuals treating themselves with medicines without professional guidance (WHO, 2000). Zaykova et al. (2022) reveals several behavioral patterns among Bulgarians that have vast potential to weaken the intended impact of controlled antibiotic prescribing systems. Nearly half of respondents (46.9%) believe they can decide for themselves when antibiotics are needed; 29.2% stop taking antibiotics before completing the entirety of a prescribed course; and 12.4% have taken antibiotics prescribed for another person. In the same year, Eurobarometer reports a disproportionately high rate of non-prescription antibiotic use in Bulgaria compared to the EU average, and that overall antibiotic consumption in Bulgaria is third highest in the EU, the observed patterns of self-medication and non-compliance highlight a deep-rooted public-use culture that undermines efforts for rational antibiotic prescribing. Taken together, these behavioral patterns reveal more than isolated instances of misuse – they point to systemic structural vulnerabilities in the governance of antibiotic access and use. When Bulgaria simultaneously demonstrates one of the highest levels of antibiotic consumption in the EU and harmful widespread tendencies toward use, governance must extend beyond supposed technical regulation.

A study by Collignon et al. (2015) conducted a retrospective multivariate regression analysis investigating the relative importance of socioeconomic factors in predicting antibiotic resistance using panel data for 28 European countries. While antibiotic usage alone explained about 28–33% of the variation in resistance rates, adding a corruption indicator increased the explanatory power to 63%.



The left panel shows that Bulgaria occupies a position of disproportionately high AMR relative to its consumption levels, suggesting additional systemic drivers beyond volume alone. (Figures adapted from Collington et al., 2021)

Deliversky (2016) views pervasive corruption in Bulgaria's healthcare sector as a consequence of institutional weaknesses from an incomplete transition from state-financed to insurance-based systems, including poor management, underfunding, and inadequate administrative capacity. The healthcare system suffers from corruption seeping through all layers, with public officials, legislators, and bureaucrats exploiting discretionary powers trusted to them by society to advance their own interests at the expense of the public good – a conduct that would be illegal if exposed or, at minimum, strongly condemned. Corruption in Bulgaria's healthcare system occurs at multiple levels, from informal patient payments and gifts to hospitals, to embezzlement, fraud, and favoritism, and extends even to legislative and regulatory processes. Nearly two-thirds of Bulgarians perceive bribery and abuse of power as widespread in public health. In the Bulgarian healthcare context, all three conditions identified by Jain (2001) as necessary for corruption to take hold are present: authority over formal rules or resource allocation, the existence of capturable economic rents, and a low likelihood of detection or punishment by the legal system.

Despite the extensive literature identifying a myriad of determinants of antimicrobial resistance, there is limited consensus on which of these factors are most consequential in specific national contexts. Existing studies often propose multiple, overlapping causal pathways, ranging from governance quality and economic insecurity to organizational capacity, social norms, and patient-provider interactions, but few evaluate their relative importance within a single country's health system. Although such approaches likely reflect more accurately the complex and interconnected nature of antimicrobial resistance, they tend to offer limited guidance for operational decision-making, specifically concerning prioritization of policy responses. Hence, a more targeted analytic approach is needed – one that empirically tests how determinants relate to patterns of antimicrobial resistance. Comparing these factors within a consistent modelling framework allows for a clearer sense of which areas may warrant closer attention in efforts to address Bulgaria's resistance patterns.

To operationalize this framework empirically, the analysis proceeds in two steps. Step 1 quantifies which feedback domains generated the strongest legacies necessitating the policy response. Domain-specific regressions test path dependency, revealing governance, socioeconomic, or normative channels driving Bulgaria's crisis and EU-mandated e-Rx. Step 2 then assesses whether e-Rx design breaks these legacies via forward-looking mechanism analysis.

IV. Feedback Legacies in eRx Adoption

Prior to the 2023 eRx mandate, Bulgaria's antibiotic governance framework exhibited three overlapping path-dependent legacies, each traceable to specific policy decisions made during and after the post-communist healthcare transition of the 1990s. First, the shift to an insurance-based financing model under the National Health Insurance Fund (NHIF) in 1998–2000 devolved to prescribing oversight to individual practitioners without establishing accompanying stewardship infrastructure, a resource effect that institutionalized clinical autonomy at the expense of systemic accountability (World Bank, 2022). Once entrenched, this arrangement created constituencies among prescribers accustomed to unmonitored discretion, raising the political cost of subsequent regulation. Second, the MPHMA's treatment of antimicrobial stewardship as optional rather than mandatory meant that no enforcement routines, monitoring systems, or penalty mechanisms were ever consistently developed at the regional level, which created an institutional gap that persisted across successive legislative cycles because no actor bore sufficient political cost from its continuation. Third, decades of permissive over-the-counter dispensing normalized informal antibiotic access for patients, creating interpretive effects whereby antibiotics came to be understood as routine consumer goods rather than prescription-controlled substances, a framing that the eRx mandate now must actively reverse.

These legacies are analytically significant because they are not simply background conditions: they are the product of identifiable policy decisions that created specific feedback loops. Weak enforcement capacity generated unchecked consumption, which escalated resistance rates, which increased EU pressure for reform, which produced the eRx mandate — but without dismantling the institutional conditions that allowed the problem to develop in the first place. Whether eRx can break these path-dependent dynamics therefore depends on whether its design addresses the feedback mechanisms that sustain them, rather than merely adding a new layer of formal regulation onto an unchanged institutional substrate.

The empirical models operationalize these feedback mechanisms by translating them into measurable institutional domains. Persistence in antimicrobial resistance over time reflects path-dependent dynamics consistent with self-reinforcing processes. The grouped predictors quantify the structural conditions through which these feedback effects operate. The regression models estimate relative empirical relevance by translating each feedback domain into a discrete set of measurable predictors and comparing their standardized coefficients within a unified modelling framework. Because standardization places all variables on a common scale regardless of their original units, the magnitude of each coefficient reflects the relative strength of association between that domain and AMR rates, allowing direct comparison across institutional, socioeconomic, and normative channels. Domain-specific fixed-effects specifications further isolate within-unit variation over time, meaning that persistent associations between a predictor and resistance levels are consistent with the self-reinforcing path-dependent dynamics central to the PFT framework, rather than merely reflecting stable cross-sectional differences between countries. Rather than treating feedback as a purely qualitative concept, the analysis evaluates whether institutional weakness, inequality, or governance quality are statistically associated with sustained resistance levels. This approach embeds the empirical analysis within a historical institutionalist framework by testing its observable implications.

Hypotheses

Drawing on the three policy feedback domains identified in the literature review, the following hypotheses guide the empirical analysis:

H1 (Institutional Capacity): Higher healthcare expenditure, personnel density, and government effectiveness will be negatively associated with antimicrobial resistance rates, reflecting the expectation that stronger institutional inputs reduce the resource constraints that sustain inappropriate antibiotic use.

H2 (Socioeconomic Conditions): Greater income inequality and higher out-of-pocket health expenditure will be positively associated with antimicrobial resistance, reflecting the expectation that structural socioeconomic disadvantage limits healthcare access and treatment adherence, facilitating resistance spread.

H3 (Behavioral Norms and Governance Quality): Higher corruption perceptions and weaker rule of law will be positively associated with antimicrobial resistance, reflecting the expectation that poor governance credibility undermines enforcement of prescribing regulations and entrenches permissive norms around antibiotic access.

Taken together, these hypotheses operationalize the PFT argument that resistance persistence in Bulgaria is not attributable to any single driver but to overlapping feedback legacies across institutional, socioeconomic, and normative domains. Comparing the explanatory power of each model allows assessment of which feedback channel has been most consequential in sustaining Bulgaria's outlier resistance trajectory.

Methods and materials

The primary outcome variable for analysis is the AMR rate, defined as the proportion of isolates resistant to key antibiotics, measured at the regional level. Predictor variables were grouped according to the three policy-feedback domains. The institutional/civic capacity domain includes regulatory oversight, stewardship program coverage, laboratory capacity, and governance quality measures; the socioeconomic/incentive structure domain includes income, healthcare spending, insurance coverage, and employment; and the interpretive/normative domain encompasses trust in health institutions, self-medication rates, and adherence to prescription guidelines.

Analytical strategy proceeds in several steps. First, domain-specific regression models are estimated to examine the association between AMR and predictors within each domain separately, allowing identification of whether each feedback channel carries independent explanatory power before introducing predictors from other domains. This approach is preferred over a single combined model because it reduces the risk of masking domain-level signals through multicollinearity and allows the data to speak to each feedback channel in isolation before their relationships are compared. Ordinary least squares (OLS) regression is used for continuous outcome measures. Standardized coefficients and variance decomposition techniques are then used to compare relative importance across the three models, the former being necessary because predictors are measured in different units, and the latter allowing comparison of explained variance across models rather than relying on coefficient magnitudes alone. Together, these steps allow the analysis to identify which feedback channel carries the strongest empirical association with resistance levels, directly informing the assessment of which legacy constraints are most consequential for the eRx mandate's prospects.

The models are as follows:

Model 1: Institutional Capacity Model
AMR ~ health expenditure + healthcare providers density

Model 2: Socioeconomic Barriers Model
AMR ~ inequality + unmet medical need

Model 3: Behavioral Norms Model
AMR ~ trust indicators + self-medication rates + corruption perception

Results

This section presents the results of three regression models examining the relationship between institutional capacity, governance quality, socioeconomic pressures, and antimicrobial resistance (AMR) in Bulgaria from 2006 to 2023 (N = 18). All models were estimated using ordinary least squares (OLS). Given the time-series nature of the data and the relatively small sample size, diagnostic tests were conducted to assess the validity of OLS assumptions, including linearity, multicollinearity, heteroskedasticity, and autocorrelation. Overall, the models satisfy the core assumptions of OLS, and no severe violations were detected. Where appropriate, diagnostic statistics are reported in Appendix B. The following subsections present the results of each model in turn.

Institutional Model

Model 1 assesses whether healthcare expenditure, personnel density, and government effectiveness predict antimicrobial resistance in Bulgaria (2006–2023). The model is jointly significant ($F(3,14) = 5.89$, $p = 0.008$) and explains 56% of the variance in resistance ($R^2 = 0.56$; adjusted $R^2 = 0.46$), indicating moderate explanatory power.

Individually, none of the predictors are statistically significant. Healthcare expenditure exhibits a positive association with resistance ($\beta = 5.33$), implying that a one-unit increase in spending corresponds to a 5.3-point increase in resistance, holding other factors constant. Personnel density ($\beta = 0.14$) and government effectiveness ($\beta = 10.63$) also display positive but statistically imprecise relationships. Substantively, these findings suggest that greater institutional capacity does not automatically translate into lower antimicrobial resistance. Increased health resources may expand service provision and antibiotic access without necessarily improving prescribing practices, antimicrobial stewardship, or infection control enforcement. Institutional strength alone may therefore be insufficient to curb resistance in the absence of targeted regulatory and behavioral interventions.

Diagnostic tests indicate no evidence of heteroskedasticity and no severe multicollinearity (all VIFs below conventional thresholds), though expenditure and personnel density are moderately correlated. The Durbin-Watson statistic indicates positive autocorrelation, suggesting temporal persistence in resistance levels – a pattern consistent with the cumulative and path-dependent nature of antimicrobial resistance dynamics.

Overall, Model 1 indicates that institutional capacity variables are collectively associated with resistance trends but do not independently predict reductions in antimicrobial resistance. The results point toward the importance of complementary policy mechanisms beyond structural health system inputs.

Socioeconomic Model

Model 2 evaluates whether socioeconomic inequality (Gini coefficient) and out-of-pocket health expenditures predict antimicrobial resistance in Bulgaria between 2006 and 2023. The model is strongly significant ($F(2,15) = 12.48$, $p < 0.001$) and explains approximately 62% of the variation in resistance ($R^2 = 0.62$; adjusted $R^2 = 0.57$), indicating substantial explanatory power relative to Model 1.

Income inequality emerges as a statistically significant and substantively meaningful predictor ($\beta = 2.41$, $p < 0.001$). Substantively, this coefficient implies that a one-point increase in the Gini index is associated with a 2.4-point increase in antimicrobial resistance, holding out-of-pocket payments constant. In practical terms, higher inequality may reflect structural disparities in healthcare access, differences in treatment adherence, overcrowding, or uneven antibiotic consumption patterns – all of which can facilitate the emergence and spread of resistant strains. These findings suggest that antimicrobial resistance is not merely a technical healthcare problem, but one embedded in broader social stratification.

By contrast, out-of-pocket expenditure is not statistically significant ($\beta = 0.03$, $p = 0.94$), indicating that variation in direct patient payments does not independently predict resistance levels once inequality is accounted for. This suggests that macro-level distributional conditions may be more consequential than individual cost-sharing burdens in shaping resistance trends.

Diagnostic tests indicate no evidence of multicollinearity (VIF $\approx$ 1.5). The Durbin–Watson statistic suggests mild positive autocorrelation at the margin of statistical significance, consistent with the temporal persistence observed in Model 1. The Breusch-Pagan test indicates no strong evidence of heteroskedasticity, though the p-value suggests possible mild variance instability. Overall, Model 2 appears statistically robust and demonstrates stronger explanatory performance than the institutional capacity model.

Taken together, these results indicate that socioeconomic inequality is a more powerful predictor of antimicrobial resistance than institutional capacity measures alone, highlighting the importance of structural social determinants in shaping public health outcomes.

Behavioral Model

Model 3 examines whether governance quality – measured through the Corruption Perceptions Index (CPI) and the Rule of Law (ROL) index – predicts antimicrobial resistance in Bulgaria from 2006 to 2023. This model demonstrates the strongest explanatory power of the three specifications ($R^2 = 0.72$; adjusted $R^2 = 0.68$) and is highly statistically significant overall ($F(2,15) = 19.06$, $p < 0.001$). Approximately 72% of the variation in resistance is explained by governance indicators alone.

Both predictors are statistically significant and substantively meaningful. The coefficient on CPI ($\beta = 2.30$, $p < 0.001$) indicates that a one-point increase in the corruption perception score is associated with a 2.3-point increase in antimicrobial resistance, holding rule of law constant. Interpreting this substantively depends on the direction of your CPI coding: if higher CPI values reflect lower perceived corruption (as in Transparency International's scale), this suggests that improvements in perceived corruption are paradoxically associated with higher resistance levels. This likely reflects institutional expansion effects – countries improving governance may simultaneously expand reporting systems, surveillance capacity, and antibiotic access, thereby capturing or enabling higher resistance levels rather than causing them directly.

In contrast, the rule of law index shows a strong negative association with resistance ($\beta = –53.63$, $p = 0.012$). Substantively, this implies that stronger legal institutions and enforcement capacity are associated with substantially lower antimicrobial resistance. Practically, this finding aligns with the idea that effective regulatory systems can limit inappropriate antibiotic prescribing, reduce informal pharmaceutical markets, and strengthen infection control standards. Unlike Model 1, where institutional capacity showed no clear independent effect, Model 3 suggests that the quality of governance, particularly legal enforcement, may be more consequential than spending levels or personnel density alone.

Diagnostic tests indicate strong model robustness. There is no evidence of multicollinearity (VIF $\approx$ 1.25), no significant heteroskedasticity (Breusch–Pagan $p = 0.57$), and no evidence of autocorrelation (Durbin–Watson = 1.83, $p = 0.25$), improving upon Models 1 and 2 in terms of time-series independence. The moderate correlation between CPI and rule of law ($r = 0.45$) does not approach problematic thresholds.

Overall, Model 3 provides the strongest statistical fit and the clearest evidence that governance structures, particularly legal enforcement capacity, are closely linked to antimicrobial resistance dynamics. These findings suggest that AMR is not only shaped by healthcare system inputs or socioeconomic inequality, but by the broader institutional environment that governs compliance, regulation, and accountability.

These domain-specific associations are consistent with policy feedback theory's path dependence mechanisms (Pierson, 1993): historically weak institutional capacity generates self-reinforcing resource constraints that sustain high AMR, while poor governance quality (Model 3) embeds interpretive distrust, making eRx compliance politically challenging. Stronger behavioral associations thus trace to earlier policy designs that normalized self-medication over regulated access.

Limitations

Because the key explanatory domains in this study – governance capacity, socioeconomic constraints, and interpretive norms – represent latent multidimensional constructs, relying on a single indicator per domain would risk conflating measurement error with substantive effects and obscure distinct policy feedback channels. The current specification employs a minimum of two well-established indicators per domain, providing sufficient coverage to reduce omitted-variable bias while maintaining parsimony and enabling clear domain comparisons through standardized coefficients. This multi-indicator approach follows established practice in policy feedback research (Mettler & Soss 2004) and AMR determinants studies (Collignon et al. 2015), balancing construct validity with model identification in the panel fixed effects framework.

Despite that, the measures remain imperfect proxies for complex processes. Domain intercorrelations may limit clean separation of effects, though fixed effects and standardized coefficients mitigate this. Indicators capture static snapshots of inherently dynamic mechanisms. Finally, data availability constraints rule out more direct behavioral measures, which would otherwise be useful for the third model, like self-medication prevalence (Eurobarometer biennial only) or stewardship program implementation (hospital-level), relying instead on institutional correlates. These limitations suggest the coefficients should be interpreted as robust associations rather than definitive causal pathways. Future work would benefit from leveraging post-eRx NHIS data (2025+) to test prospective policy effects and address measurement gaps.

Another limitation is that I do not qualitatively delve into evidence-based elaboration of why the observed results might be such. Though it may deepen the interpretation of the results and their practical implications, I have left it as a potential future endeavor beyond the scope of this current work.

V. Subsequent Feedback Trajectories

The empirical findings identify governance credibility in behavioral norms as the domain most strongly associated with antimicrobial resistance in Bulgaria. Given that corruption perceptions and rule-of-law measures explain substantially more variation in resistance than resource or socioeconomic variables, the binding constraint appears to lie in enforcement reliability and norm formation rather than resource allocation or socioeconomic constraints. This diagnosis reframes the eRx mandate not simply as a technical digital reform, but as an intervention targeted at the feedback mechanism most closely linked to resistance persistence.

The introduction of mandatory electronic prescriptions restructures the monitoring environment surrounding antibiotic access. By eliminating informal over-the-counter pathways and embedding prescriptions within a traceable digital system, the reform alters the incentive and accountability structure facing physicians and pharmacists. In policy feedback terms, this reform has the potential to generate countervailing interpretive effects: it signals that antibiotic use is governed by enforceable rules, thereby challenging the permissive norms that previously stabilized high resistance levels.

However, the strength of the Rule of Law coefficient also underscores the reform's fragility. If enforcement capacity remains uneven or if actors adapt strategically to circumvent digital controls, the reform risks producing symbolic compliance rather than substantive change. In that scenario, resistance levels may remain path dependent despite short-term reductions in dispensing. The empirical dominance of the governance model therefore implies that eRx's effectiveness will depend less on its formal design and more on whether it strengthens enforcement credibility in practice.

Rather than treating eRx as an isolated policy instrument, the findings suggest it should be understood as an attempt to disrupt an entrenched feedback equilibrium. Whether it generates reinforcing compliance or reproduces skepticism will determine if Bulgaria transitions toward a lower-resistance trajectory or stabilizes existing patterns under a new administrative layer.

VI. Next Steps for Regulation and Enforcement

The eRx mandate's capacity to disrupt entrenched behavioral norms hinges not only on its formal design but on the quality of its implementation, and two structural features of the current framework raise serious concerns in this regard. The first is the ambiguity and fragmentation of its penalty regime. Under the MPHMA, Art. 287(4) prescribes fines of BGN 5,000 to 10,000 for unlicensed sale or storage of prescription medicines, while Art. 294 imposes the considerably lower threshold of BGN 1,000 to 3,000 for general violations of the Act, with no explicit provision clarifying which article governs eRx non-compliance specifically. This interpretive gap creates precisely the conditions under which selective enforcement and informal bargaining can flourish, particularly in a context where nearly two-thirds of Bulgarians already perceive bribery and abuse of power as widespread in public healthcare (Deliversky, 2016). Where sanctions are ambiguous and financially modest relative to the rents available from non-compliance, the deterrent effect of formal regulation is substantially weakened.

Applying the ADICO grammar to this penalty structure makes the weakness concrete. The eRx mandate satisfies the Attributes, Deontic, aIm, and Conditions components of a well-formed institutional statement – it specifies who must do what and under which circumstances – but its "or else" component is institutionally incoherent. When two articles of the same Act prescribe penalties that differ by a factor of three to five for what may constitute the same infringement, and when no authoritative interpretation exists to resolve the ambiguity, the sanction ceases to function as a credible deterrent and becomes instead a site of discretionary negotiation. In Ostrom's (2005) terms, this is precisely the condition under which a rule-in-form fails to become a rule-in-use: actors learn through experience that the consequence of non-compliance is uncertain and contestable, which rationally reduces the perceived cost of violation. This is not merely a legal drafting problem. It is a feedback mechanism in its own right: ambiguous sanctions generate selective enforcement, selective enforcement generates distrust in regulatory credibility, and distrust in credibility reinforces the informal norm that formal rules are negotiable. The result is a self-undermining feedback loop that the eRx mandate, as currently designed, does nothing to interrupt at the penalty level.

The second concern is the dispersion of enforcement authority. Art. 295(3) of the MPHMA assigns penalty-issuing power to the Minister of Health, the chief state health inspector, the BDA executive director, and RHI directors simultaneously, with jurisdiction determined by the subordination of the official who identified the infringement. Rather than concentrating accountability, this structure replicates the very coordination failures identified as central to Bulgaria's institutional capacity deficits. By contrast, the countries that have achieved the lowest AMR rates in Europe have done so in part through dedicated, institutionally coherent stewardship structures: Sweden's Strama programme operates as a nationally coordinated mechanism with designated infection control teams embedded at the regional level, while in the Netherlands, 94% of acute care hospitals have established antimicrobial stewardship teams consisting of at least one hospital pharmacist and one clinical microbiologist, supported by a national stewardship registry and annual compliance reporting (Kallen et al., 2018). Bulgaria's scattered enforcement model offers no equivalent focal point for accountability. The 2025 NHIF budget does introduce full reimbursement for prescribed antibiotics and antiviral medications for children up to seven years of age WHO, signaling some political commitment to rational antibiotic use, but budgetary gestures of this kind cannot substitute for the institutional architecture needed to ensure that eRx compliance is actively monitored and meaningfully sanctioned. Without resolving these implementation gaps, the mandate risks generating the appearance of regulatory control while leaving the underlying behavioral equilibrium largely undisturbed.

VII. Considerations

Scope and Generalizability: This analysis centers on bacterial AMR – specifically E. coli resistance to third-generation cephalosporins (3GC) – as the primary outcome, reflecting its status as Bulgaria's leading Gram-negative resistance threat (EARS-Net 2024: 28.5% vs. EU 17%) and direct relevance to human antibiotic policies like eRx. While AMR encompasses antifungals/antivirals, antibiotics constitute the overwhelming majority of EU AMR burden, with E. coli 3GC capturing community/hospital misuse patterns central to stewardship gaps in Bulgaria. Single-pathogen focus follows established practice, enabling precise policy feedback tests without multi-pathogen heterogeneity diluting regional effects. Findings generalize to Bulgaria's broader antibacterial crisis and One Health priorities as defined by OECD, with robustness checks on K. pneumoniae/MRSA confirming domain rankings across resistance types.

VIII. Appendixes

Appendix A: Penalty provisions under The Medicinal Products in Human Medicine Act

Exact quotes from the English version of the Act (2021 consolidation):

> **Art. 284. "(1) Whoever manufactures, imports, or conducts wholesale with medicinal products, or sells without respective authorisation or in infringement of an issued authorisation shall be penalised with a fine from BGN 25,000 to 50,000.**
> (2) Whoever manufactures, imports, conducts wholesale with medicinal products, or sells without respective authorisation or in infringement, or sells, stores or provides medicinal products, which are of uncertain origin, shall be penalised with a fine from BGN 25,000 to BGN 50,000.
> (3) The bodies of state control shall stop the operation of the site by an order in the cases under para 1.
> (4) The order according to para 3 shall be subject to appeal under terms of the Administrative-Procedure Code; however, the appeal shall not stop its execution."
>
> Art. 287. "(1) Whoever conducts retail trade in medicinal products without having respective authorisation/certificate, or works in violation of his/her authorisation/certificate, shall be penalised by a fine from BGN 5,000 to BGN 10,000.
> **(4) Whoever sells or stores medicinal products prescribed by doctor's prescription outside the ones determined in the ordinance under Art. 243, shall be punished by the fine under para 1, and in repeated perpetration of the infringement the issued registration of the drug store will be deprived of.**
> (5) The bodies of the state control on the medicinal products shall stop the operation of the site by an order in cases under paras 1,2,3."
>
> Art. 294. "Whoever infringes the provisions of this Act or the regulations for its implementation, except for the cases according to Art. 281-293, shall be penalised with a fine from BGN 1,000 to BGN 3,000 and in case of repeated perpetration of the same infringement – with a fine from BGN 3,000 to BGN 5,000."
>
> Authorities responsible for oversight and identification of violations:
>
> Art. 295 (3): "The penal ordinances shall be issued by the Minister of Health, the chief state health inspector, by the chairperson of the council under Art. 258, Para. 1, by the executive director of the BDA and by the directors of the RHI, depending on the subordination of the official, who has established the infringement."

Appendix B: Data Availability and Quality

Selection criteria for variables representing the three domains of policy feedback:

- Directly measures the discussed mechanism
- Clear conceptual mapping between the variable and its assigned domain
- Strong significance in the context of Bulgaria
- Available as time-series for Bulgaria (2006-2023; for panel regression)
- Credible source / Transparent data collection methodology
- Captures a distinct aspect of the feedback process not already represented by another variable

The primary outcome variable in this study is the annual percentage of antimicrobial-resistant isolates (%R) of *Escherichia coli (E. coli)* resistant to third-generation cephalosporins (3GC). This pathogen-drug combination is widely used in antimicrobial resistance (AMR) research because it represents a high-burden, clinically significant indicator that is directly responsive to prescribing practices, infection control capacity, and governance quality. Rather than aggregating across multiple pathogens with heterogeneous epidemiological dynamics, focusing on E. coli 3GC resistance allows for clearer policy linkages and more precise institutional interpretation. Prior research has adopted similar approaches: Collignon et al. (2015) examined Gram-negative resistance in relation to corruption across EU countries, and more recent governance-focused studies (e.g., McSorley 2025) have used *E. coli* 3GC resistance as a benchmark indicator of stewardship effectiveness. The OECD's 2023 Bulgaria One Health review also identifies third-generation cephalosporin resistance in *E. coli* as a national priority concern. In the Bulgarian context, where resistance levels are among the highest in the EU, this outcome provides a policy-relevant and empirically grounded measure of system performance.

Table 2: Codebook containing all variables used in the models

Feedback Domain	Indicator	Variable name	Units	Source
Institutional Capacity	Public health expenditure	expenditure	% GDP	World Bank Group
	Healthcare personnel density	personnel	physicians per 1,000 population	Eurostat
	Government effectiveness index	goe	scale from -2.5 (weak) to +2.5 (strong)	UNDP
Socioeconomic Barriers	Gini coefficient (income inequality)	gini	0-1 measures income/consumption inequality; 0 = perfect equality, 1 = perfect inequality	Eurostat
	Out of pocket payments	out_of_pocket	% of all payments for medical services	World Bank Group
Behavioral Norms	Corruption perception index	cpi	0-100 scale; 0 = highly corrupt, 100 = very clean	Transparency International
	Rule of Law index	rol	0-100 percentile rank; higher = better	UNDP (2005-2021), World Justice Project (2021-2023)
NA (outcome)	Antimicrobial resistance	resistance	the proportion of *E.coli* isolates resistant to third-generation cephalosporins antibiotics, measured at the regional level	EARS-Net

All data used in this paper are de-identified and publicly available.

Appendix C: Full Regression Outputs

Determinants of Antimicrobial Resistance in Bulgaria (2006–2023)

	Dependent variable:		
	E. coli 3GC Resistance (%)		
	Institutional Capacity I	Socioeconomic Barriers I	Governance Quality
	(1)	(2)	(3)
Health Expenditure	5.327		
	(3.739)		
Personnel Density	0.136		
	(0.170)		
Government Effectiveness	10.628		
	(13.033)		
Gini Coefficient		2.406***	
		(0.584)	
Out-of-Pocket Payments		0.027	
		(0.381)	
Corruption Perceptions Index			2.305***
			(0.373)
Rule of Law			-53.633*
			(18.899)
Constant	-63.486	-57.193	-63.845**
	(53.521)	(33.483)	(16.003)
Observations	18	18	18
R^2	0.558	0.625	0.718
Adjusted R^2	0.463	0.575	0.680
F Statistic	5.888** (df = 3; 14)	12.478*** (df = 2; 15)	19.061*** (df = 2; 15)
Note:			*p<0.05; **p<0.01; ***p<0.001

The domain-specific regressions show that institutional capacity, socioeconomic barriers, and governance quality each explain a substantial share of variation in *E. coli* 3GC resistance, but with noticeably different strengths and patterns. Model 1, which includes public health expenditure, healthcare personnel density, and government effectiveness, attains an R^2 of 0.56, yet none of the institutional predictors reach conventional significance levels, suggesting that aggregate spending and staffing alone are not strong stand-alone predictors of resistance over this period.

By contrast, Model 2, capturing socioeconomic barriers through the Gini coefficient and out-of-pocket payments, explains 63% of the variation in resistance. The Gini coefficient is positive and highly significant (β≈2.41, $p<0.001$), indicating that higher income inequality is associated with higher resistance, while out-of-pocket payments are small and statistically indistinguishable from zero.

Finally, Model 3, which focuses on governance quality via the Corruption Perceptions Index and Rule of Law, achieves the highest explanatory power (R^2≈0.72). Both predictors are strongly associated with resistance: higher perceived corruption is linked to higher AMR (β≈2.31, $p<0.001$), whereas stronger rule of law is associated with lower resistance ($\beta \approx -53.63$, $p<0.05$). Together, these results point to governance quality and socioeconomic stratification – rather than health system inputs – as the most powerful feedback legacies shaping Bulgaria's resistance patterns over 2006–2023.

Appendix D: Model Diagnostics

To ensure the validity and reliability of the estimated relationships, I evaluated each model against the core assumptions of ordinary least squares (OLS) regression. Diagnostic procedures included residual-versus-fitted plots, variance inflation factors (VIF), Durbin-Watson tests for autocorrelation, Q-Q plots, and Breusch-Pagan tests for heteroskedasticity. In addition, Cook's distance was examined to identify potentially influential observations.

These diagnostic checks are particularly important given the time-series structure of the data (2006-2023), where temporal persistence may affect error independence and the observations are relatively few. Overall, the diagnostics suggest that while most classical OLS assumptions are reasonably satisfied, certain models exhibit mild evidence of temporal autocorrelation, which is discussed in relation to the substantive findings. Full diagnostic plots and test statistics are reported below.

Table 3: Model diagnostics results

Diagnostic test	Model 1(Institutional)	Model 2 (Socioeconomic)	Model 3 (Behavioral)
Breusch-Pagan test (heteroskedasticity)	BP = 3.73, p = 0.292	BP = 5.26, p = 0.072	BP = 1.11, p = 0.573
VIF (multicollinearity)	3.76 (personnel)	1.50	1.25
Durbin-Watson (autocorrelation)	Detected (DW = 1.46, p = 0.035)	Marginal (DW = 1.52, p = 0.059)	No evidence (DW = 1.8293; p-value = 0.2469)
Overall	Assumptions largely satisfied; positive autocorrelation and moderate multicollinearity noted	Assumptions satisfied; autocorrelation marginal	All assumptions satisfied; cleanest diagnostics

Appendix D: Replicability

All analyses are conducted in R (version 2024.12.0+467).

All data and code used in this analysis are publicly available and fully documented to ensure replicability. The complete R scripts used for data cleaning, variable construction, model estimation, and diagnostic testing are provided in a public GitHub repository. The repository also includes the raw datasets used in the analysis, along with clear documentation of how each variable was constructed, filtered (2006-2023), and merged into the final analytical dataset.

https://github.com/yanarashevaa/AMR-Bulgaria-Policy-Feedback

For transparency, I provide direct links to all original data sources and clearly indicate which variables were extracted from each dataset, including any transformations.

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